Techniques for Dynamic Visualization of Time-Varying Flood Data

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December 11, 2025

Abstract

This project focuses on creating clear and accessible visualizations of flood events using satellite data from the Copernicus Global Flood Monitoring system. Centered on the 2018 Thessaly flood in Greece, the goal is to represent flood extent and likelihood in a way that is easily understood by the general public. Methods include literature review, visualization design, and user evaluation with an emphasis on interpretability and accessibility.

1 Introduction

The Global Flood Monitoring (GFM) information from Copernicus Emergency Management Services (CEMS) [12] provided for this project consists of satellite-observed time-dependent flood data in the geographical region of Thessaly, Greece. These data reflect the extent, intensity, and uncertainty of the flood over time. More precisely, we will focus on the flood event at the end of February, 2018 [5], which was caused by intense high rainfall over a short period of time. Visualizing these changes in a way that is comprehensible to a broader audience, like the general public, presents a unique challenge.

2 Problem Statement

The goal of this project is to design and present new visualization techniques that effectively represent time-dependent flood events, ensuring that they are easily accessible and understandable to the general public. The research question is:

How can flood risk be effectively combined and visualized for the general public?

Our definition of flood risk is a combination of current and future flood likelihood and flood extent. We define the general public as individuals without prior experience in analyzing flood visualizations who still need to accurately interpret flood risk information, even in unfamiliar areas. For our evaluation, we recruited participants without formal training in geography, flood monitoring, or data science. The sample did include adults aged 18 to 65 years, including individuals without a university degree. Our target users are not expected to have in-depth knowledge of the regions under assessment.

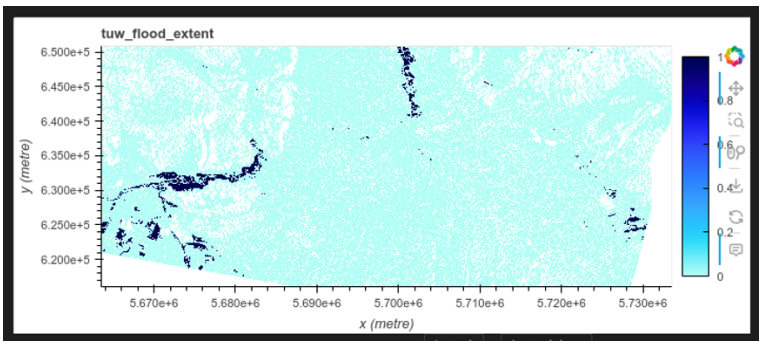


Figure 1: Basic flood extent visualization

3 Methodology

3.1 Literature Research

The first step was to gain a deeper understanding of flood visualizations and to draw inspiration from visualization techniques. The literature review focuses on papers published within the last 10 years that address the visualization of flood risk (including flood likelihood and flood extent) for non-expert audiences.

Therefore, existing literature and studies on geospatial data visualization, such as satellite-based flood mapping (2017)[1]. This study focuses on flood mapping through bayesian inference, but still has some interesting visualizations as inspiration in it.
Color map design for visualization in flood risk assessment was the topic of the research of Seipel et al. in their work [9]. They use a sequential color scale for displaying uncertainty of the flooding status. They also use color blue as indication for flooded areas.
The use of polygons to represent information on flood maps or to support flood-related analysis is well established in the literature. For example, some studies employ deep learning to map flood extents and visualize the resulting water

areas as polygons [7], while others use polygon-based water surfaces to support flood-depth estimation and related calculations [2].
Rahman et al. (2018) propose overlaying flood maps from different timestamps to highlight temporal changes in flood extent. This technique allows several days of flood information to be presented in a single image. This can also hand itself to show which areas remain flooded the longest [8]. Although we did not apply this method directly, we use a similar technique for an approach in section 6.
The research of Cohen et al. focuses on estimating floodwater depths [3]. In their approach, smoothing techniques are applied to reduce abrupt, unrealistic variations in water depth, resulting in more gradual and visually coherent transitions between affected areas.
The rather current work of Zhu et al. [13] from 2025 used simulated 3D flood imagery as flood maps. This representation should help to improve risk cognition and risk preparedness. Therefore, they generate short video of a 3D flood simulation of a street view. This did increase the risk cognition and generated stronger emotional responses.

3.2 Data & Pre-processing

This project utilizes the Global Flood Monitoring dataset, which offers flood detection data based on satellite observations. The data set, primarily pre-processed, focuses on the Thessaly region in Greece. For each coordinate and time point, it provides estimates of flood likelihood and extent from multiple models, along with a mask indicating permanent bodies of water.
Although the data set was largely preprocessed, further refinement was necessary to make it suitable for public-facing visualizations. This pre-processing steps were of small scope, like removing NaN-values for some visualizations, or binning the data for a specific array.

One pre-processing step that needed to be mentioned extra, was the re-projection of the spatial dimensions. The x and y coordinates given in the data were in a projected coordinate system, which uses linear units (meters). To use it with other geographic datasets and visualization on standard maps, the x and y coordinates needed to be using longitude and latitude in degrees. This was done be re-projecting the data to WGS84 (longitude/latitude). Finally, the re-projected data is clipped to a specific geographic extent defined by minimum and maximum longitude and latitude.

3.3 Visualizations

The idea was to implement new visualization techniques, which could then be compared in terms of effectiveness to the baseline presented in Figure 1. The baseline visualization provides limited information about the affected region and its surroundings. To improve interpretability, we added a background map to the baseline visualization. Since background maps were included in all the more

refined visualizations, omitting it in the baseline would have made comparisons disproportionately difficult. By adding the map, we created a secondary baseline that is more comparable, ensuring that a crucial piece of context is not missing.

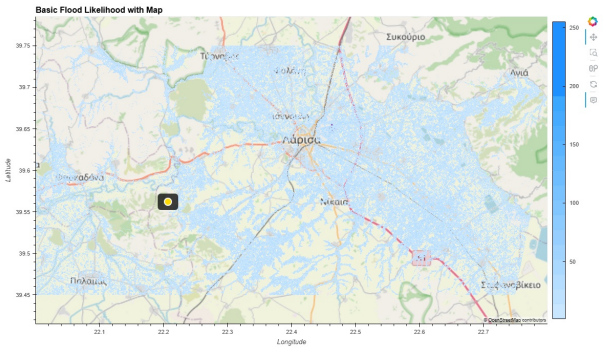


Figure 2: Baseline visualization with background map

In the next approach, we introduced polygon-based visualizations to smooth the underlying flood-likelihood values. Instead of showing pixel-level variations, the data was aggregated into larger, contiguous polygon areas. We intentionally generated broader and more connected polygons to produce only a few distinct regions, allowing users to form quick and easy interpretations of the patterns. This smoothing reduces visual fragmentation and creates more uniform areas, making spatial trends easier to recognize. The edges of the polygons are much more defined, which helps in estimating the extent of the likelihood area.

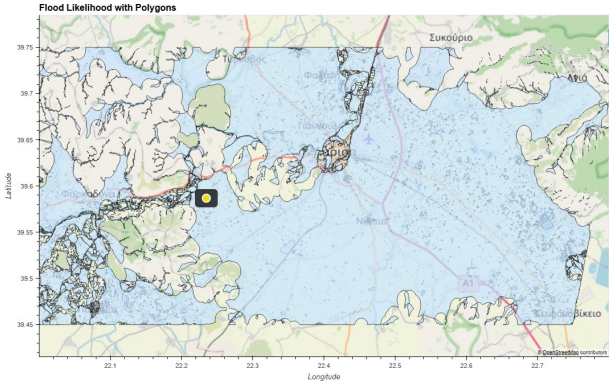


Figure 3: Flood likelihood smoothing with polygons

In the mixed visualization approach, we combined polygon smoothing with the original raster information and the flood extent. In this method, the smooth-

ing is minimal and primarily serves to make the boundaries of the areas more visible and create uniform areas of likelihood by reducing small holes within larger high-likelihood zones. The underlying pixel-level data is partially retained to preserve finer detail where it is relevant. The now included flood extent provides much additional information. Users can perceive general flood-likelihood patterns through the polygons while still seeing the more dangerous areas within each region. The mixed approach therefore provides a middle ground between a highly smoothed visualization and a fully detailed raster view.

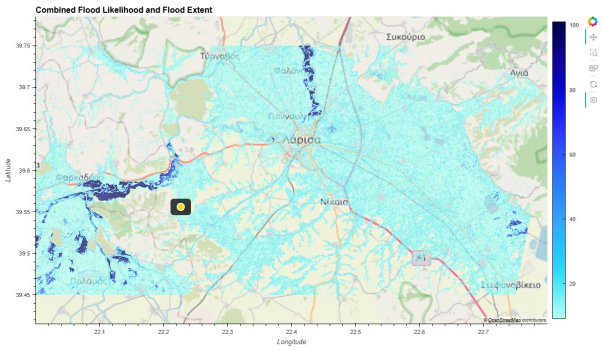


Figure 4: Flood likelihood and extent with minimal smoothing

Finally, the binning-based visualization further refines the polygon concept by grouping areas into discrete flood-likelihood classes. Instead of continuous values, the data is segmented into bins that represent meaningful risk categories. Each bin is assigned a distinct visual encoding, allowing users to quickly distinguish between levels of flood likelihood. This method should help simplifying the value space and emphasizes the categorical differences that are often more useful in decision-making contexts. The binning visualization therefore supports given better likelihood information to the users.

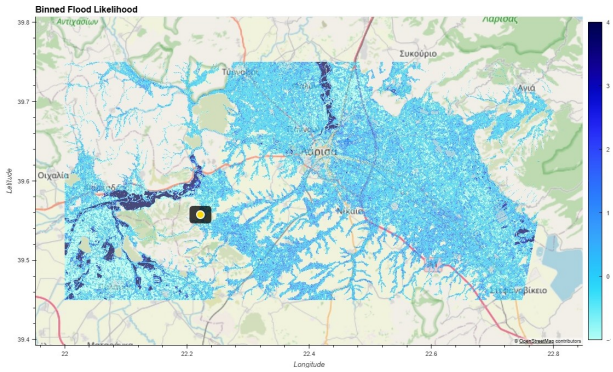


Figure 5: Binned flood likelihood

The primary goal was to create visualizations that are both intuitive and informative, with a particular focus on accessibility for non-technical users and the broader public. Consequently, approaches involving interactive dashboards or time sliders were dismissed, as they were considered overly complex for the task.

4 Survey

The survey consisted of three parts and it was conducted using SoSci Survey, an online survey platform [6], which allows both the creation and hosting of surveys. The target number of participants was 30 completed responses, which was successfully achieved.
The target audience of the survey was very broad, as it should display the general public. Therefore, the participant group should also not consist mainly of data science students, as they would have an above average understanding of visualizations and how to read them.

The first part of the survey, is simple to get a basic information of the participant and their prior knowledge of the survey topic. This helps to determine the general public's understanding of the subject and can also be loosely compared to their confidence in later questions. This page appears always first as asks the following question with these possible answers: *How familiar are you with reading flood maps or geospatial maps?*

- Not familiar at all
- Slightly familiar
- Moderately familiar
- Very familiar
- Expert

Next comes the main part of the survey, in which the participants are given questions to the different visualizations. This section spans several pages, with each page about one visualization. The order of these pages was randomized, to not give any visualization an advantage. This was very important because the inexperienced user is probably more unsure about the first visualization and more confident after already seeing multiple flood visualizations of the area. The pages are structured so that the participants will complete a simple task question to the visualization and then a second question assessing their confidence in their answer. For example, see the figure 6 at the end of the report.
The first question is always *Based on this visualization, is it currently safe for a person entering from the south to travel to the yellow point shown on the map?* with the possible answer 'Yes', 'No', and 'Unsure'. The point is placed in a location where the right answer should theocratically always be 'No'. This would

allows us to see if the participants can read the visualization correctly.

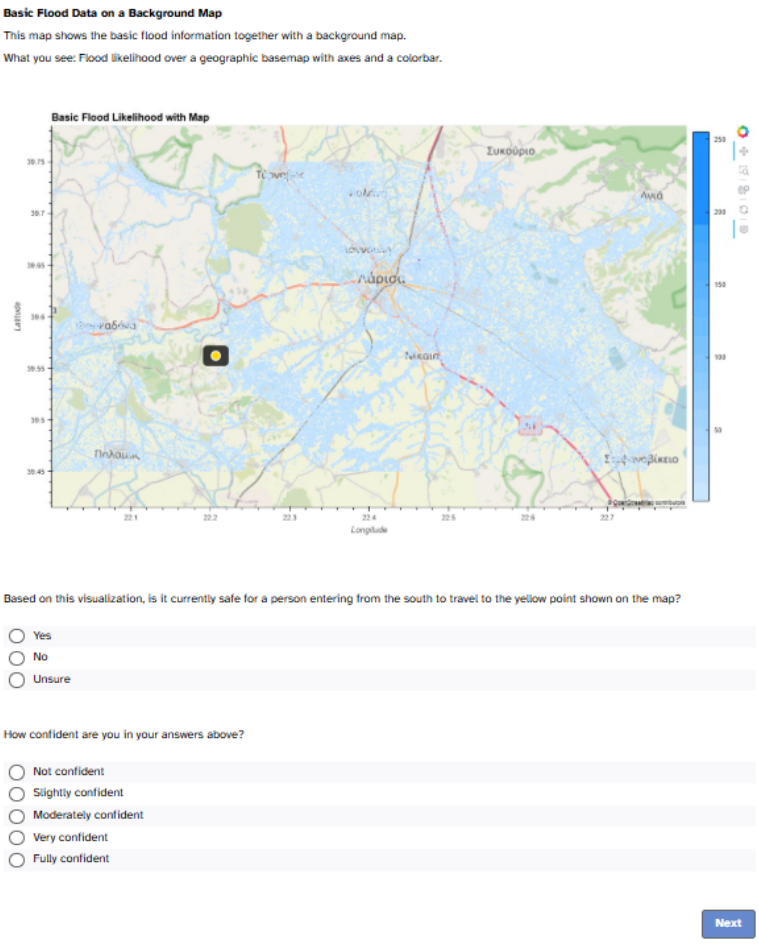


Figure 6: Survey page

The final part of the survey asked participants to rank the different visualizations based on *Which visualization was the easiest for you to read and understand?*. This page was always placed at the end of the survey to ensure that participants had already seen each visualization at least once. The page displayed all visualizations again as reference examples, allowing participants to revisit them if needed. At the bottom, participants were instructed to drag and drop the names of the visualizations along a scale ranging from "easiest" (left) to "hardest" (right).
The survey software returned values between 0 and 1000 for each visualization, based on where it was positioned on the scale. These values can be easily converted into a ranking of the visualizations.

Despite the careful design of the survey, several limitations should be acknowledged. Many of these became more apparent after receiving feedback from some participants.
First, the task question should have differed more clearly between visualizations. After answering the first few questions, participants likely formed an expectation about the "correct" answer and continued to provide the same response throughout. As a result, the later questions may not have provided meaningful information about the interpretability of those visualizations.
Second, the survey was designed and tested on a computer with a large screen. However, it is very likely that most participants completed the survey on mobile devices, as the link was distributed via WhatsApp. This would clearly affect the visibility and interpretability of the visualizations, as they are displayed much smaller on mobile screens. The biggest downside of using a mobile device was in the last question, where the participants had to rank the visualizations. On the small screen participants had to scroll up to view the example visualizations and then scroll back down to make the ranking by drag-and-drop the names of the visualizations in the preferred order. During this process, it was easy to accidentally reset some already ranked visualizations. This could probably lead to participants just ordering them of the top of their head without consulting the example visualizations.

5 Results

The effectiveness of the proposed visualizations was evaluated, focusing mainly on interpretability and accessibility. Therefore, user feedback was collected through a testing survey to determine whether the new visualizations provide improvements over the baseline visualizations developed by TU Wien. Each of the multiple visualization techniques that were implemented, was subjected to the same evaluation procedure.
Overall, the differences in the results between the visualizations were not as large as expected. The baseline visualizations performed worse in some aspects of the survey, but in other areas they showed performance comparable to the other visualizations.

From the first question of the survey we can extract the familiarity of the participants with the topic. Only the three lowest familiarity options were selected, with the majority indicating that they were "not familiar at all." This outcome was expected from a sample group that intended to represent the general public.

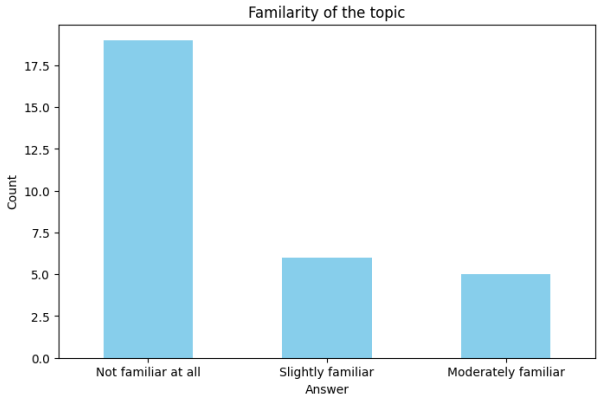


Figure 7: Familiarity

The main part of the survey shows us results for two questions. The task-question for the visualization and secondly the confidence of the answer.
Looking at the results of the task question, we can clearly see in Fig.8 that the baseline visualization received the answer "Unsure" significantly more often than the other visualizations. This suggests that it was more difficult for participants to interpret. On the other hand, the number of "No" responses ,which represents the correct answer, is similar across all visualization types. This indicates that all visualizations were prone to being misunderstood to some extent.

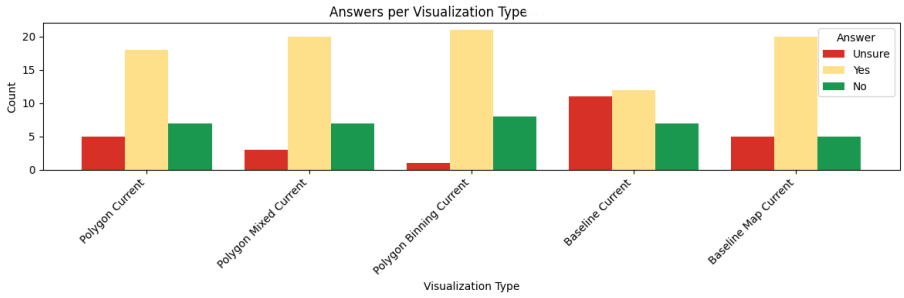


Figure 8: Answer to the task question

The confidence ratings collected in the second question confirm that more participants reported being "not confident" in their answers for the baseline and baseline-with-map visualizations, see Fig.9. This aligns with the increased number of "Unsure" responses observed in the first question. Then again, looking at the other answer possibilities, no clear trend is visible, that the baseline is worse. In fact, the baseline with map even got the most answers with 'Fully confident'.
If the average confidence is calculated by assigning numerical values from 0 ("not

confident") to 4 ("fully confident"), the baseline visualization performs worst, with an average score of around 1.3. The remaining visualizations achieve average confidence levels between approximately 1.6 and 1.7. The baseline-with-map visualization stands out as an exception, reaching an average confidence above 2, largely due to the higher number of "fully confident" responses it received.

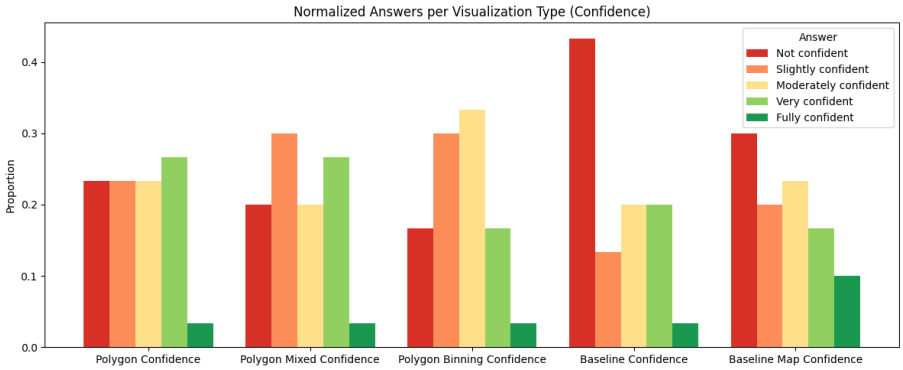


Figure 9: Confidence in the answers

The ranking results from the final page of the survey are particularly interesting. They show that almost all visualizations received a very similar average rank of around 3, even the normal baseline. The only notable outlier is again the baseline with map visualization, but this time it performs worse with an average rank of around 3.4 , see Fig.10. This suggests that, overall all visualizations are almost similar difficult to read, with the exception of the baseline with map. This outcome appears somewhat counterintuitive given that the baseline-with-map performed better in terms of confidence ratings.

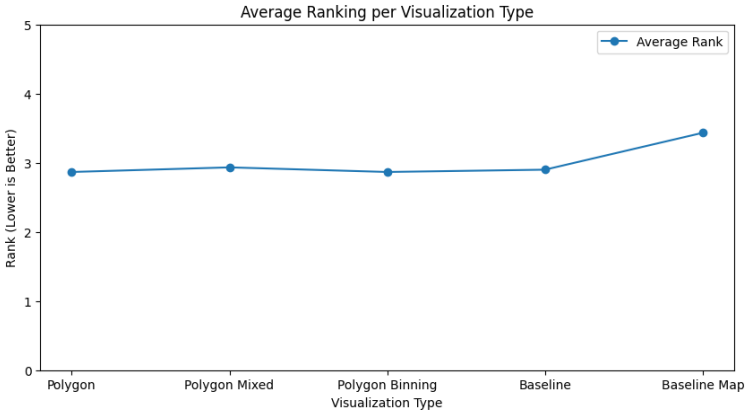


Figure 10: Average ranking of the visualizations

The survey also provided the time measurements of how long a participants did stay on each page. Looking at the average page time of each visualization the range from 29 seconds on the binning visualization page to 18 seconds on the normal polygon and the mixed visualization page. Both baseline pages lie in the middle with 22 seconds visiting time. There is no direct connection visible, between visiting time of a page and the ranking, confidence, or answering performance.

5.1 Conclusion & Discussion

After reviewing all results, it is difficult to conclude definitively that the baseline visualization performs significantly worse than the other visualization types. While participants appeared less confident when answering questions related to the baseline, reflected in the higher frequency of "Unsure" responses, this pattern is not repeated in the overall ranking of the visualizations. In fact, the final ranking does not indicate that the baseline is perceived as harder to read.
Overall, presenting flood visualizations to unprepared and inexperienced participants appears to be challenging, regardless of the visualization type. The results indicate that participants were slightly more confident when using the non-baseline visualizations compared to the baseline, but still have a hard time answering questions.

Several points can be raised when discussing the results.
First, as already noted in the survey section, the final page containing the ranking task was particularly difficult to operate on mobile devices. This may have distorted the responses in this part of the survey.
Second, the survey was intentionally designed as a fully blind survey, targeting participants with no prior knowledge of the topic. While this approach helped represent the general public, it also meant that many participants may have simply guessed their answers. This is reflected in the generally low confidence ratings throughout the survey. Including an additional introductory page that explained the procedure in more detail, supported by example visualizations, could have significantly improved participants understanding and potentially led to more reliable responses.
A valuable next step would be to conduct a second survey in which participants with a similar baseline knowledge are thoroughly briefed on the survey process and expected questions. The results of this informed survey could then be compared to those of the current study to assess the impact of prior knowledge and guidance.

5.2 Accessibility

Accessibility was a key consideration in the creation of the visualizations. Special attention was paid to color selection to ensure support for individuals with color vision deficiencies. The visualizations were tested using Coblis, a Color

Blindness Simulator [4].
Each visualization was examined under different types of color vision deficiency to assess interpretability. The colors remained distinguishable in all forms of color vision deficiency, although in some cases they were harder to differentiate, except in the case of *Monochromacy/Achromatopsia*, where all colors appear as shades of gray, representing total color blindness [10].

For accessibility, we also considered the Web Content Accessibility Guidelines (WCAG), which are international standards established by the World Wide Web Consortium (W3C) to ensure accessible content on the internet [11].
Although these guidelines were not directly applicable to our visualizations in this context, they would be highly relevant if the visualizations were deployed on the web in a real-world scenario. Adhering to WCAG would help guarantee that the information is accessible to the widest possible audience.

6 Supplementary Considerations

In addition to creating the visualizations and conducting the survey, several extra steps were carried out that fall outside the scope of the main project. These steps represent potential improvements or the basics for a follow-up project. They are included here as examples of further work and investigation.

Firstly, because the Global Flood Monitoring (GFM) data is captured by a satellite moving along its orbital path, the data it selects for a specific time point does not need to have the perfect shape of the area we want to analyses. As a result, a single time step does not necessarily have the full extent of the map of interest. This is visible in the left visualization in Figure 11, where NaN-values are shown in dark blue: on the left side, the sea is excluded as expected, but on the right side the image contains only NaN-values, clearly revealing a cut, where the scanning area of the satellite ends for this timestamp.
To address this issue and produce maps without missing areas, a seamless flood-likelihood composite is created using a First-Valid-Pixel temporal mosaic. Basically, script starts with the earliest image and iteratively fills its missing pixels using valid data from later swaths of the satellite. Each subsequent time slice contributes only where the data still has gaps, and the process continues until either all missing pixels are filled or no more images remain. This process generates visualizations like the right one in Figure 11.
This does only make sense, if the times of satellite scans are not far apart, otherwise the visualization could lose its meaning. However, when the scans are taken within a short interval, it could be a good idea for giving an overview of how conditions in the area have changed in a day.

Secondly, there is the idea of generating 3D visualizations that combine flood depth with the terrain elevation of the region. Such a presentation could be very useful for predicting how a flood might spread and for estimating the impact it

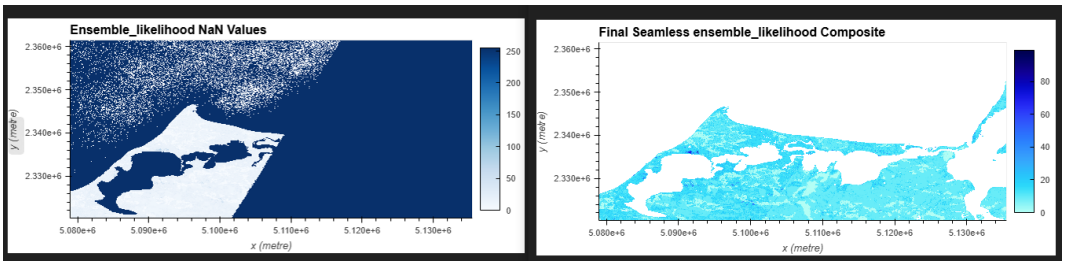


Figure 11: NaN values of original data vs. gap-free composite of the area

has on an already affected areas.
To generate this type of visualization, both the flood depth and the elevation of the area need to be known. In principle, this information could be obtained from elevation models and water-level sensors. This kind of map would valuable for the general public, as it shows safe locations (areas of higher elevation) and regions that should be avoided. It could also support decision-makers by indicating zones that are just outside the current flood extent and are likely to be affected next, to do having low-lying terrain adjacent to already flooded areas. An example of what this visualization might look like is shown in Figure 12. This approach was tested only with simulated data, as the available real-world data was missing some of the information required to implement it.

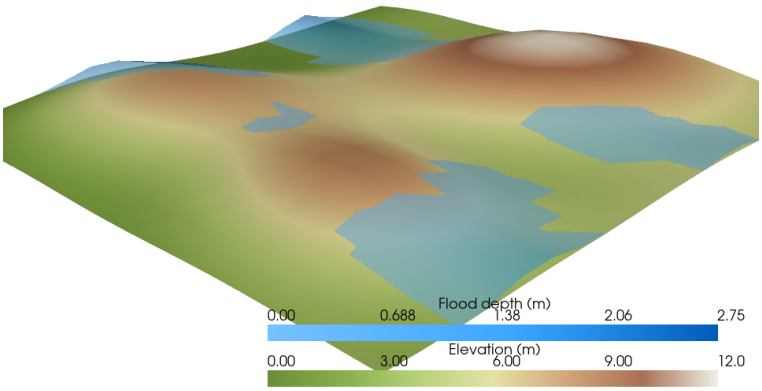


Figure 12: 3D visualizations with flood depth and terrain elevation

7 Domain-specific lecture

I attended the domain-specific lecture '120.110 Data Retrieval in Earth Observation' and completed it on 17.06.2025.

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